CSCI 576 Multimedia Project

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The course project is meant to give you an in depth understanding of some of the areas in multimedia technology. Since this is a broad field, there can be a variety interesting projects that can be done depending on your interests which can also extend to related and complementary topics that are taught in class.

Also, I have often found that a large project can be successfully accomplished via collaboration. Additionally, working together to design and integrate code can be a rewarding exercise and you will frequently need to work in teams when you set out to work in the industry. Accordingly, please form groups of exactly two - three students. Please use Piazza to form groups by posting about your interests and availability. You are free to form groups with in-class and remote DEN students together.

This time I want to suggest a topic that involves removing specific, sometimes unwanted content from audio video data. On the next page is a complete description giving the motivation and the requirements of your project. It should be a comprehensive project that aims to increase your fundamental understanding of rendering video and audio in a synchronized manner and analyzing the multimodal signals with topics discussed in class.

Multimodal Segmentation of Long-Form Online Video into Content and Non-Content

There is an increased amount of video and audio content recorded, posted, hosted everywhere today from your personal recordings to availability on social media platforms like YouTube, Instagram, X etc. Online video is now one of the dominant forms of communication, entertainment, education, and advertising. But long-form video often contains material that viewers do not value equally: intros, outros, sponsorships, self-promotion, repeated recaps, "starting soon" screens, dead air, transition screens, unrelated insertions, and other non-core segments. These segments can interrupt the viewing experience, reduce accessibility, and make large video archives harder to navigate.

The goal of this project is to build a **multimodal system** that automatically segments (or classifies) long-form videos into:

- core content
- non-content segments

where non-content may include one or more of the following: sponsorships, ads, intros, outros, recaps, transition cards, holding screens, filler, dead space, or other material that is not central to the primary purpose of the video.

Your task is **not** merely to classify individual frames or transcript snippets. Instead, you will design and implement an end-to-end system that analyzes video, audio, speech, text, and visual structure to identify meaningful temporal segments and produce useful outputs for end users. Along with the segmentation/classification you will also need to develop a player that can play the video (with synchronized audio) showing a "visual map" that will allow you to appropriately navigate and skip/jump content that you deem not to waste your time on. For example, there may be ads that once they appear you may want to skip, or if a movie, you may want to skip the preamble showing cast names, and other times you may want to consume them depending on your interest in the content.

Your system should answer questions like the following, and your custom video player should allow you to navigate the non-content (skip, fast forward, play as is etc.) as a choice:

- Where does the actual content begin?
- Which intervals are likely sponsorships or self-promotional interruptions?
- Which parts are repeated across episodes?
- Which intervals contain little informational or entertainment value?

Think about how such a qualitative and important "content map" can be presented to the user.

The final deliverable should be a practical tool that can be run offline to generate metadata which you can use in your visual player while consuming. For example, your system might generate some combination of these elements put together:

- a timeline with labeled segments

- chapter markers
- skip suggestions
- summaries of segment types
- a cleaned playback interface
- a browser-like review tool for human verification

The emphasis is on **multimodal reasoning** that creates metadata or a "content map" which in turns allows you to build a practically useful video player. This problem space addresses a real multimedia problem where a good solution needs to detect "media semantics" like visual patterns, motion and shot structure, music and sound, timing and repetition etc. A robust project solution should provide a genuine value-added proposition by improving video browsing/navigation experience, the media accessibility/indexability, media search, ultimately increasing watch-time efficiency,

Why will this project be difficult in the right way?
There is no single clean definition or even signal for what "non-content" defines. A sponsorship segment may look visually similar to the main video, while an intro may be obvious visually but not aurally or linguistically. Some segments are repetitive, some are subtle, and some depend on context. This makes the problem rich enough and potential solutions varied enough depending on how much thought you and your group put into defining "non-content". To be able to effectively demonstrate your solution, your process need to not only generate reasonably accurate results, but your player/system should be able to showcase your results by offering a flexibility of how to easily and seamlessly navigate through the video depending on what a user might find/feel "interesting" enough to watch and consume.

What Counts as "Non-Content"
While the exact definition is bound to vary, your team must define an operational taxonomy and for this you may use a binary setup or a more detailed one. A possible taxonomy could be proposed as :

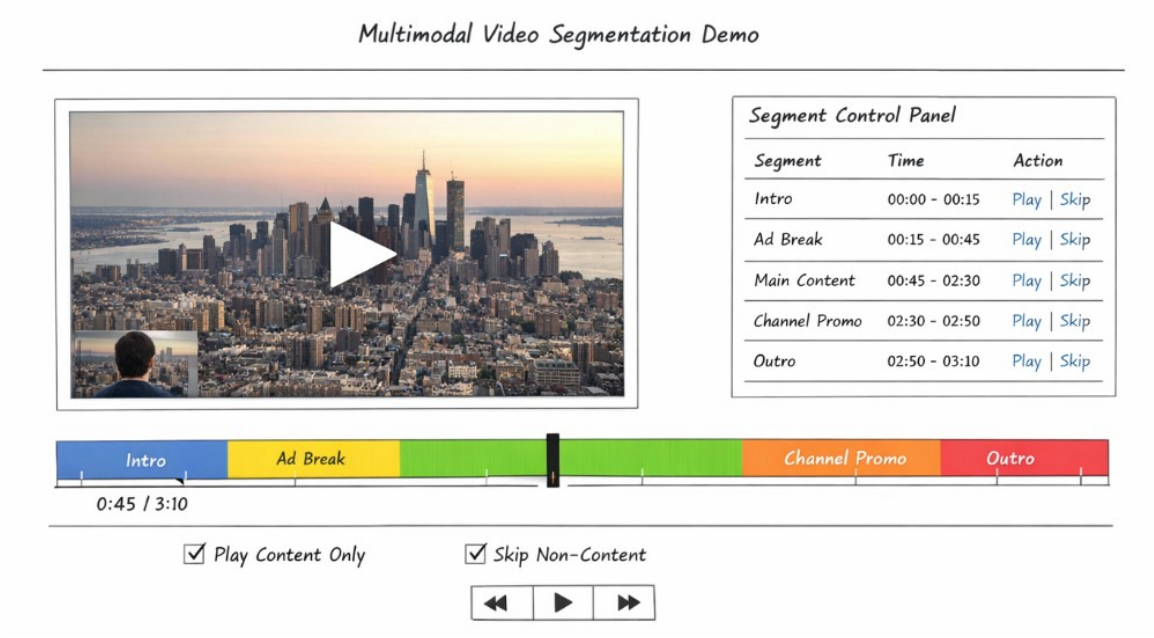
- **Core content**
  the main material the viewer wants to watch in a video
- **Non-content**
  - intro
  - outro
  - sponsorship / advertisement
  - self-promotion / channel promotion
  - recap / repeated boilerplate
  - transition / intermission screen
  - dead air / silence / inactivity
  - waiting room / countdown / holding screen
  - unrelated insert or filler

You are allowed to simplify this if needed, but your definitions must be explicit and consistently applied

Required Project Goals

Your system must do the following:

1. Ingest long-form video that could educational lectures, podcasts, gaming streams, interviews, talk shows, movies, news clips
2. Use multiple modalities in your analysis and figure out motion / shot structure. Multiple modalities might include - audio frequency analysis, visual frame color structure, motion between frames, speech transcripts, etc.
3. A player that can represent segmented temporal segments of the video along with navigation capabilities – play, stop, skip to next segment, with an interactive timeline. Some example interfaces are included below for illustrative purposes only)


Multimodal Video Segmentation Demo
Segment Control Panel
Segment
Time
Action
Intro
00:00 - 00:15
Play | Skip
Ad Break
00:15 - 00:45
Play | Skip
Main Content
00:45 - 02:30
Play | Skip
Channel Promo
02:30 - 02:50
Play | Skip
Outro
02:50 - 03:10
Play | Skip
Intro
Ad Break
Channel Promo
Outro
0:45 / 3:10
Play Content Only
Skip Non-Content

4.